RAMAS ACADEMY

Class 12 · Board Exam Preparation · OMR Test Series

Class 12 Chemistry – OMR Chapter Test – Detailed Answer Key

NCERT Chemistry Part I (2026–27) · Unit 5: Coordination Compounds · based on CBSE board questions (2020–2025)

Time Allowed: 3 Hours | **Total Questions: 60** | **Maximum Marks: 80**

Quick Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	C	D	B	A	C	B	A	D	D	B	C	D	C	B	A	A	C	B	D

21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
C	D	A	B	A	B	C	D	D	A	B	C	A	C	D	B	C	A	B	D

41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
D	C	B	A	A	D	B	C	C	B	D	A	A	D	C	B	D	C	B	A

Scoring: correct = full marks of the section; wrong or multiple bubbles = −25% of the question's marks; blank = 0.

Section A

1-mark level (incl. Assertion–Reason) · 33 × 1 = 33 marks · −0.25 per wrong

Q1. According to Werner's theory, the primary valency of a metal ion in a complex corresponds to its: [1]

Werner's Theory & Key Terms *CBSE PYQ*

(A) oxidation state **(B)** number of ligands

(C) number of unpaired electrons **(D)** coordination number

Answer: (A)

1. Primary valency = ionisable valency = oxidation state; secondary valency = coordination number

Q2. One mole of $CoCl_3 \cdot 4NH_3$ reacts with excess $AgNO_3$ to give 1 mol of AgCl. Its formula is: [1]

Werner's Theory & Key Terms *CBSE PYQ*

(A) $[Co(NH_3)_4]Cl_3$ **(B)** $[Co(NH_3)_4Cl]Cl_2$

(C) $\mathbf{[Co(NH_3)_4Cl_2]Cl}$ **(D)** $[Co(NH_3)_3Cl_3] \cdot NH_3$

Answer: (C)

1. Only one Cl^- is outside the coordination sphere ⇒ $[Co(NH_3)_4Cl_2]Cl$ (CN = 6)

Q3. The coordination number and oxidation state of Fe in $[Fe(C_2O_4)_3]^{3-}$ are: Werner's Theory & Key Terms *CBSE PYQ* [1]

(A) 3 and +6 **(B)** 3 and +3 **(C)** 6 and +2 **(D) 6 and +3**

Answer: (D)

1. Oxalate is bidentate: 3 × 2 = 6; x + 3(−2) = −3 ⇒ x = +3

Q4. $EDTA^{4-}$ is a: Werner's Theory & Key Terms *CBSE PYQ* [1]

(A) tetradentate ligand **(B) hexadentate ligand**

(C) monodentate ligand **(D)** bidentate ligand

Answer: (B)

1. EDTA binds through 2 N and 4 O donor atoms ⇒ hexadentate

Q5. Which of the following is an ambidentate ligand? Werner's Theory & Key Terms *CBSE PYQ* [1]

(A) $\mathbf{NO_2^-}$ **(B)** NH_3 **(C)** H_2O **(D)** en

Answer: (A)

1. NO_2^- can bind through N (nitrito-N) or O (nitrito-O)

Q6. The number of ions produced in solution by one formula unit of $[Co(NH_3)_6]Cl_3$ is: Werner's Theory & Key Terms [1]

CBSE PYQ

(A) 3 **(B)** 9 **(C) 4** **(D)** 6

Answer: (C)

1. $[Co(NH_3)_6]^{3+} + 3Cl^- = 4$ ions

Q7. A ligand that binds to a metal through two donor atoms forming a ring is called a: Werner's Theory & Key Terms [1]

CBSE PYQ

(A) monodentate ligand **(B) chelating ligand**

(C) ambidentate ligand **(D)** bridging ligand only

Answer: (B)

1. A di- or polydentate ligand forming a ring (e.g. en, $C_2O_4^{2-}$) is a chelating ligand

Q8. **Assertion (A):** Chelate complexes are more stable than similar complexes with monodentate ligands. [1]

Reason (R): Chelating ligands form rings with the metal atom (chelate effect). Werner's Theory & Key Terms *CBSE PYQ (A–R)*

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is not the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Answer: (A)

1. Ring formation by polydentate ligands increases stability (chelate effect) ⇒ A true and R explains it ⇒ option (A)

Q9. The IUPAC name of $[Co(NH_3)_5Cl]Cl_2$ is: IUPAC Nomenclature *CBSE PYQ* [1]

(A) pentaamminechloridocobaltate(III) chloride **(B)** pentaamminechloridocobalt(II) chloride

(C) chloridopentaamminecobalt(III) chloride **(D) pentaamminechloridocobalt(III) chloride**

Answer: (D)

1. Ligands in alphabetical order: ammine before chlorido; Co: x + 0 − 1 = +2 ⇒ x = +3; cationic complex ⇒ no '-ate'

Q10. The IUPAC name of $K_3[Fe(CN)_6]$ is: IUPAC Nomenclature *CBSE PYQ* [1]

(A) tripotassium hexacyanoiron(III) **(B)** potassium hexacyanidoiron(III)

(C) potassium hexacyanidoferrate(II) **(D) potassium hexacyanidoferrate(III)**

Answer: (D)

1. Anionic complex ⇒ ferrate; 3(+1) + x + 6(−1) = 0 ⇒ Fe = +3

Q11. The formula of tetraamminediaquacobalt(III) chloride is: IUPAC Nomenclature *CBSE PYQ* [1]

(A) $[Co(NH_3)_4(H_2O)_2]Cl$ **(B) $[Co(NH_3)_4(H_2O)_2]Cl_3$**

(C) $[Co(NH_3)_4(H_2O)_2]Cl_2$ **(D)** $[Co(NH_3)_4(H_2O)Cl]Cl_2$

Answer: (B)

1. Co(III) with neutral ligands ⇒ charge +3 ⇒ three Cl^-

Q12. The IUPAC name of $[Pt(NH_3)_2Cl_2]$ is: IUPAC Nomenclature *CBSE PYQ* [1]

(A) dichloridodiammineplatinum(II) **(B)** diamminedichloridoplatinum(IV)

(C) diamminedichloridoplatinum(II) **(D)** diamminedichloridoplatinate(II)

Answer: (C)

1. Neutral complex: ammine (a) before chlorido (c); Pt: x − 2 = 0 ⇒ +2

Q13. The oxidation state of nickel in $[Ni(CO)_4]$ is: IUPAC Nomenclature *CBSE PYQ* [1]

(A) +2 **(B)** −2 **(C)** +4 **(D) 0**

Answer: (D)
1. CO is neutral and the complex is neutral ⇒ Ni = 0

Q14. The IUPAC name of $K_2[Zn(OH)_4]$ is: IUPAC Nomenclature *CBSE PYQ* [1]

(A) potassium tetrahydroxidozincate(IV) **(B)** dipotassium tetrahydroxozinc(IV)
(C) potassium tetrahydroxidozincate(II) **(D)** potassium tetrahydroxidozinc(II)

Answer: (C)
1. Anionic ⇒ zincate; 2 + x − 4 = 0 ⇒ Zn = +2

Q15. The IUPAC name of $[Cr(en)_3]Cl_3$ is: IUPAC Nomenclature *CBSE PYQ* [1]

(A) tri(ethylenediamine)chromate(III) chloride **(B) tris(ethane-1,2-diamine)chromium(III) chloride**
(C) triethylenediaminechromium(III) trichloride **(D)** tris(ethane-1,2-diamine)chromium(II) chloride

Answer: (B)
1. For ligand names that already contain numbers, use bis/tris with brackets; Cr = +3

Q16. In the IUPAC name of a complex anion, the name of the central metal ends in: IUPAC Nomenclature *CBSE PYQ* [1]

(A) -ate **(B)** -ide **(C)** -ite **(D)** -ium

Answer: (A)
1. e.g. ferrate, cobaltate, zincate for anionic complexes

Q17. **Assertion (A):** The IUPAC name of $[Ni(CN)_4]^{2-}$ is tetracyanidonickelate(II) ion. [1]
Reason (R): In an anionic complex, the name of the metal ends with the suffix '-ate'. IUPAC Nomenclature *CBSE PYQ (A–R)*

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

Answer: (A)
1. x + 4(−1) = −2 ⇒ Ni = +2; anionic ⇒ nickelate ⇒ A true, R explains it ⇒ option (A)

Q18. $[Co(NH_3)_5(NO_2)]Cl_2$ and $[Co(NH_3)_5(ONO)]Cl_2$ show: Isomerism *CBSE PYQ* [1]

(A) geometrical isomerism **(B)** coordination isomerism
(C) linkage isomerism **(D)** ionisation isomerism

Answer: (C)
1. The ambidentate NO_2^- is bonded through N in one and O in the other

Q19. $[Co(NH_3)_5SO_4]Br$ and $[Co(NH_3)_5Br]SO_4$ are examples of: Isomerism *CBSE PYQ* [1]

(A) hydrate isomers **(B) ionisation isomers**
(C) optical isomers **(D)** linkage isomers

Answer: (B)
1. They give different ions in solution (Br^- vs SO_4^{2-})

Q20. $[Cr(H_2O)_6]Cl_3$ (violet) and $[Cr(H_2O)_5Cl]Cl_2{\cdot}H_2O$ (grey-green) show: Isomerism *CBSE PYQ* [1]

(A) coordination isomerism **(B)** linkage isomerism
(C) optical isomerism **(D) solvate (hydrate) isomerism**

Answer: (D)
1. Water is inside the coordination sphere in one and outside in the other

Q21. $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$ are: Isomerism *CBSE PYQ* [1]

(A) ionisation isomers
(B) geometrical isomers
(C) coordination isomers
(D) linkage isomers

Answer: (C)
1. Ligands are interchanged between the cationic and anionic complexes

Q22. A square planar complex $[MA_2B_2]$ can show: Isomerism *CBSE PYQ* [1]

(A) both geometrical and optical isomerism
(B) no isomerism
(C) optical isomerism
(D) geometrical (cis-trans) isomerism

Answer: (D)
1. cis and trans forms exist; the planar molecule has a mirror plane, so it is optically inactive

Q23. The number of geometrical isomers of $[Co(NH_3)_3(NO_2)_3]$ (octahedral MA_3B_3) is: Isomerism *CBSE PYQ* [1]

(A) 2 (fac and mer) **(B)** none **(C)** 3 **(D)** 4

Answer: (A)
1. MA_3B_3 shows facial (fac) and meridional (mer) isomers

Q24. Which of the following complexes is optically active? Isomerism *CBSE PYQ* [1]

(A) trans-$[Pt(NH_3)_2Cl_2]$
(B) cis-$[Co(en)_2Cl_2]^+$
(C) trans-$[Co(en)_2Cl_2]^+$
(D) $[Co(NH_3)_6]^{3+}$

Answer: (B)
1. The cis form has no plane of symmetry ⇒ non-superimposable mirror images; the trans form has a mirror plane

Q25. **Assertion (A):** Tetrahedral complexes do not show geometrical isomerism. [1]
Reason (R): In a tetrahedral complex, the relative positions of the ligands attached to the central atom are the same with respect to each other. Isomerism *CBSE PYQ (A–R)*

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

Answer: (A)
1. All positions in a tetrahedron are adjacent to one another, so cis/trans arrangements are impossible ⇒ A true and R explains it ⇒ option (A)

Q26. The hybridisation and magnetic nature of $[Ni(CN)_4]^{2-}$ are: Bonding (VBT & CFT), Colour & Magnetism *CBSE PYQ* [1]

(A) dsp^2, paramagnetic
(B) dsp^2, diamagnetic
(C) sp^3, paramagnetic
(D) sp^3, diamagnetic

Answer: (B)
1. Ni^{2+} ($3d^8$): strong-field CN^- pairs the electrons, freeing one 3d orbital ⇒ dsp^2, square planar, no unpaired electrons

Q27. The hybridisation and shape of $[NiCl_4]^{2-}$ are: Bonding (VBT & CFT), Colour & Magnetism *CBSE PYQ* [1]

(A) dsp^2, square planar
(B) d^2sp^3, octahedral
(C) sp^3, tetrahedral
(D) sp^3d^2, octahedral

Answer: (C)
1. Cl^- is weak field: no pairing in $3d^8$; 4s and 4p used ⇒ sp^3, tetrahedral, 2 unpaired electrons

Q28. $[Co(NH_3)_6]^{3+}$ is an inner orbital complex while $[CoF_6]^{3-}$ is an outer orbital complex. The number of unpaired electrons in $[CoF_6]^{3-}$ is: Bonding (VBT & CFT), Colour & Magnetism *CBSE PYQ* [1]

(A) 2 **(B)** 0 **(C)** 3 **(D) 4**

Answer: (D)

1. Co^{3+} $3d^6$; F^- is weak field ⇒ high spin $t_{2g}^4 e_g^2$ ⇒ 4 unpaired (sp^3d^2)

Q29. In an octahedral crystal field, the d orbitals split into: Bonding (VBT & CFT), Colour & Magnetism *CBSE PYQ* [1]

(A) lower e_g (two) and higher t_{2g} (three) **(B)** five degenerate orbitals

(C) lower e (two) and higher t_2 (three) **(D) lower t_{2g} (three) and higher e_g (two)**

Answer: (D)

1. Ligands approach along the axes, raising $d_{x^2-y^2}$ and d_{z^2} (e_g)

Common trap: Lower e and higher t_2 is the tetrahedral pattern.

Q30. Which ligand produces the strongest crystal field splitting? Bonding (VBT & CFT), Colour & Magnetism *CBSE PYQ* [1]

(A) CN^- **(B)** H_2O **(C)** F^- **(D)** Cl^-

Answer: (A)

1. Spectrochemical series: $I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 < en < CN^- < CO$

Q31. The spin-only magnetic moment of $[Fe(CN)_6]^{3-}$ is about: Bonding (VBT & CFT), Colour & Magnetism *CBSE PYQ* [1]

(A) 5.92 BM **(B) 1.73 BM** **(C)** 0 BM **(D)** 2.83 BM

Answer: (B)

1. Fe^{3+} $3d^5$; strong-field CN^- ⇒ low spin t_{2g}^5 ⇒ 1 unpaired ⇒ $\mu = \sqrt{3} \approx 1.73$ BM

Common trap: 5.92 BM would be the high-spin value.

Q32. Anhydrous $CuSO_4$ is white but $CuSO_4 \cdot 5H_2O$ is blue because: Bonding (VBT & CFT), Colour & Magnetism *CBSE PYQ* [1]

(A) the anhydrous salt absorbs all visible light

(B) anhydrous $CuSO_4$ has no d electrons

(C) in the anhydrous salt there are no water ligands to cause d–d splitting

(D) water reduces Cu^{2+} to Cu^+

Answer: (C)

1. Without ligands, no crystal field splitting occurs, so no d–d transition absorbs visible light ⇒ white

Q33. **Assertion (A):** $[Ti(H_2O)_6]^{3+}$ is coloured. [1]

Reason (R): Ti^{3+} has one d electron that can undergo a d–d transition by absorbing visible light.

Bonding (VBT & CFT), Colour & Magnetism *CBSE PYQ (A–R)*

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is not the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Answer: (A)

1. Ti^{3+} ($3d^1$) absorbs light (~498 nm) for $t_{2g} \rightarrow e_g$ ⇒ violet colour ⇒ A true, R explains it ⇒ option (A)

Section B

2-mark level (Very Short Answer type) · 6 × 2 = 12 marks · −0.5 per wrong

Q34. When 0.1 mol of $CoCl_3(NH_3)_5$ is treated with excess $AgNO_3$, 0.2 mol of AgCl is obtained. The conductivity of its solution will correspond to: [2]

Werner's Theory & Key Terms · *CBSE PYQ (NCERT)*

(A) a 1 : 1 electrolyte
(B) a 1 : 3 electrolyte
(C) a 1 : 2 electrolyte
(D) a non-electrolyte

Answer: (C)

1. 2 Cl^- per formula are ionisable ⇒ $[Co(NH_3)_5Cl]Cl_2 \rightarrow [Co(NH_3)_5Cl]^{2+} + 2Cl^-$
2. One cation and two anions ⇒ 1 : 2 electrolyte

Q35. The correct formula of potassium tri(oxalato)chromate(III) and of hexaamminecobalt(III) sulphate are: [2]

IUPAC Nomenclature · *CBSE PYQ*

(A) $K_2[Cr(C_2O_4)_3]$ and $[Co(NH_3)_6]SO_4$
(B) $K[Cr(C_2O_4)_3]$ and $[Co(NH_3)_6]_3(SO_4)_2$
(C) $K_3[Cr(C_2O_4)_3]$ and $[Co(NH_3)_6](SO_4)_3$
(D) $K_3[Cr(C_2O_4)_3]$ and $[Co(NH_3)_6]_2(SO_4)_3$

Answer: (D)

1. $[Cr(C_2O_4)_3]$: +3 + 3(−2) = −3 ⇒ needs 3 K^+
2. $[Co(NH_3)_6]^{3+}$ with SO_4^{2-}: 2 × (+3) = 3 × (−2) ⇒ $[Co(NH_3)_6]_2(SO_4)_3$

Q36. The total number of isomers (geometrical + optical) of $[Co(en)_2Cl_2]^+$ is: [2]

Isomerism · *CBSE PYQ*

(A) 4 **(B) 3** **(C)** 1 **(D)** 2

Answer: (B)

1. Two geometrical isomers: cis and trans
2. The cis isomer is chiral and exists as a pair of enantiomers (d and l); trans is achiral
3. Total = 2 (cis d, l) + 1 (trans) = 3

Q37. How many geometrical isomers are possible for the square planar complex $[Pt(NH_3)(Br)(Cl)(py)]$? [2]

Isomerism · *CBSE PYQ (NCERT)*

(A) 0 **(B)** 4 **(C) 3** **(D)** 2

Answer: (C)

1. For square planar MABCD, three geometrical isomers are possible (fix A; B, C or D can be trans to it)

Q38. Using CFT, the electronic configuration of d^4 ion in an octahedral field when $\Delta_o > P$ (pairing energy) is: [2]

Bonding (VBT & CFT), Colour & Magnetism · *CBSE PYQ*

(A) $t_{2g}^4 e_g^0$ **(B)** $e_g^4 t_{2g}^0$ **(C)** $t_{2g}^3 e_g^1$ **(D)** $t_{2g}^2 e_g^2$

Answer: (A)

1. When $\Delta_o > P$, the 4th electron pairs up in t_{2g} (low spin)
2. ⇒ $t_{2g}^4 e_g^0$; if $\Delta_o < P$ it would be $t_{2g}^3 e_g^1$

Q39. $[Fe(H_2O)_6]^{3+}$ is strongly paramagnetic whereas $[Fe(CN)_6]^{3-}$ is weakly paramagnetic. The numbers of unpaired electrons are respectively: [2]

Bonding (VBT & CFT), Colour & Magnetism · *CBSE PYQ (NCERT)*

(A) 1 and 5 **(B) 5 and 1** **(C)** 3 and 1 **(D)** 5 and 3

Answer: (B)

1. Fe^{3+} $3d^5$: H_2O (weak) ⇒ high spin, 5 unpaired
2. CN^- (strong) ⇒ low spin t_{2g}^5, 1 unpaired

Section C

3-mark level (Short Answer type) · 3 × 3 = 9 marks · −0.75 per wrong

Q40. Write the IUPAC names of: (i) $[Co(NH_3)_4(H_2O)Cl]Cl_2$, (ii) $[Ni(CO)_4]$, (iii) $K_3[Cr(C_2O_4)_3]$. The correct set is: [3]

IUPAC Nomenclature *CBSE PYQ*

(A) (i) tetraammineaquachloridocobalt(III) chloride; (ii) nickel tetracarbonyl(II); (iii) tripotassium chromium oxalate

(B) (i) aquatetraamminechloridocobalt(III) chloride; (ii) tetracarbonylnickelate(0); (iii) potassium trioxalatochromate(III)

(C) (i) tetraammineaquachloridocobalt(II) chloride; (ii) tetracarbonylnickel(II); (iii) potassium trioxalatochromium(III)

(D) (i) tetraammineaquachloridocobalt(III) chloride; (ii) tetracarbonylnickel(0); (iii) potassium trioxalatochromate(III)

Answer: (D)

1. (i) Ligands alphabetically: ammine, aqua, chlorido; Co: x − 1 = +2 ⇒ +3
2. (ii) Neutral CO ligands ⇒ Ni(0)
3. (iii) Anionic ⇒ chromate; Cr: x − 6 = −3 ⇒ +3

Q41. Using valence bond theory, predict the hybridisation, shape and magnetic behaviour of $[Co(NH_3)_6]^{3+}$ (Co: Z = 27). [3]

Bonding (VBT & CFT), Colour & Magnetism *CBSE PYQ*

(A) sp^3, tetrahedral, diamagnetic

(B) d^2sp^3, octahedral, paramagnetic

(C) sp^3d^2, octahedral, paramagnetic

(D) d^2sp^3, octahedral, diamagnetic

Answer: (D)

1. Co^{3+}: $[Ar]3d^6$; NH_3 causes pairing ⇒ three filled 3d orbitals, two empty 3d orbitals
2. Two 3d + one 4s + three 4p ⇒ d^2sp^3 (inner orbital), octahedral
3. All electrons paired ⇒ diamagnetic

Q42. Calculate the crystal field stabilisation energy (CFSE) of a d^6 ion in a strong-field (low-spin) octahedral complex, ignoring pairing energy. [3]

Bonding (VBT & CFT), Colour & Magnetism *CBSE pattern*

(A) 0 **(B)** $-0.4\ \Delta_o$ **(C) $-2.4\ \Delta_o$** **(D)** $-1.6\ \Delta_o$

Answer: (C)

1. Low spin d^6: $t_{2g}^6 e_g^0$
2. Each t_{2g} electron = $-0.4\Delta_o$, each e_g = $+0.6\Delta_o$
3. CFSE = $6(-0.4) + 0 = -2.4\Delta_o$

Common trap: $-0.4\Delta_o$ is the high-spin ($t_{2g}^4 e_g^2$) value.

Section D

4-mark level (Case Studies – 4 MCQs per case) · 16 × 1 = 16 marks · −0.25 per wrong

Case Study 1: Werner's Cobalt Ammines

Werner's Theory & Key Terms

Werner prepared complexes of composition $CoCl_3 \cdot 6NH_3$ (yellow), $CoCl_3 \cdot 5NH_3$ (purple) and $CoCl_3 \cdot 4NH_3$ (green and violet forms). He treated each with excess $AgNO_3$ solution.

Answer Q43–46 on the basis of the above information.

Q43. The moles of AgCl precipitated per mole of $CoCl_3 \cdot 6NH_3$ are: [1]

(A) 0 **(B) 3** **(C)** 1 **(D)** 2

Answer: (B)

1. $[Co(NH_3)_6]Cl_3$: all three Cl^- are ionisable

Q44. The formula of the purple complex is: [1]

(A) $[Co(NH_3)_5Cl]Cl_2$ **(B)** $[Co(NH_3)_6]Cl_3$

(C) $[Co(NH_3)_5]Cl_3$ **(D)** $[Co(NH_3)_5Cl_2]Cl$

Answer: (A)

1. CN = 6 ⇒ 5 NH_3 + 1 Cl inside; 2 Cl^- outside

Q45. The green and violet forms of $CoCl_3 \cdot 4NH_3$ are: [1]

(A) trans and cis geometrical isomers **(B)** coordination isomers

(C) linkage isomers **(D)** ionisation isomers

Answer: (A)

1. $[Co(NH_3)_4Cl_2]Cl$ exists as trans (green) and cis (violet)

Q46. The secondary valency of cobalt in all these complexes is: [1]

(A) 4 **(B)** 3 **(C)** 5 **(D) 6**

Answer: (D)

1. Secondary valency = coordination number = 6

Case Study 2: Cisplatin Isomerism

cis-$[Pt(NH_3)_2Cl_2]$ (cisplatin) is an anticancer drug; its trans isomer is not effective.

Answer Q47–50 on the basis of the above information.

Q47. The geometry of the complex is: [1]

(A) octahedral **(B) square planar**

(C) tetrahedral **(D)** trigonal bipyramidal

Answer: (B)

1. Pt(II), d^8 with dsp^2 ⇒ square planar

Q48. The oxidation state of Pt is: [1]

(A) 0 **(B)** +4 **(C) +2** **(D)** +3

Answer: (C)

1. x + 0 + 2(−1) = 0 ⇒ +2

Q49. The type of isomerism shown by $[Pt(NH_3)_2Cl_2]$ is: [1]

(A) linkage **(B)** optical **(C) geometrical** **(D)** ionisation

Answer: (C)

1. MA_2B_2 square planar ⇒ cis/trans

Q50. Cisplatin is optically: [1]

(A) inactive, because it is tetrahedral **(B) inactive, because it has a plane of symmetry**

(C) active, because Pt is chiral **(D)** active, because it is cis

Answer: (B)

1. The flat molecule is its own mirror image ⇒ optically inactive

Case Study 3: Colours of Nickel Complexes Bonding (VBT & CFT), Colour & Magnetism

$[Ni(H_2O)_6]^{2+}$ is green, $[Ni(NH_3)_6]^{2+}$ is blue-violet and $[Ni(CN)_4]^{2-}$ is colourless/yellow. Ni: Z = 28.

Answer Q51–54 on the basis of the above information.

Q51. The number of unpaired electrons in $[Ni(H_2O)_6]^{2+}$ is: [1]

(A) 0 **(B)** 1 **(C)** 4 **(D) 2**

Answer: (D)

1. Ni^{2+} $3d^8$ in octahedral field: $t_{2g}^6 e_g^2$ ⇒ 2 unpaired

Q52. The spin-only magnetic moment of $[Ni(NH_3)_6]^{2+}$ is: [1]

(A) 2.83 BM **(B)** 4.90 BM **(C)** 1.73 BM **(D)** 0 BM

Answer: (A)
1. n = 2 ⇒ √8 = 2.83 BM

Q53. Replacing H_2O by NH_3 shifts the absorbed light to a shorter wavelength because NH_3: [1]

(A) is a stronger field ligand, increasing Δ_o
(B) is a weaker field ligand
(C) reduces Ni^{2+}
(D) has no lone pair

Answer: (A)
1. Larger Δ_o ⇒ higher energy (shorter λ) absorbed ⇒ different complementary colour

Q54. $[Ni(CN)_4]^{2-}$ is: [1]

(A) square planar and paramagnetic **(B)** octahedral and diamagnetic
(C) tetrahedral and paramagnetic **(D) square planar and diamagnetic**

Answer: (D)
1. Strong-field CN^- ⇒ dsp^2, all electrons paired

Case Study 4: Crystal Field in Iron Complexes Bonding (VBT & CFT), Colour & Magnetism

The pairing energy P for Fe^{2+} is about 210 kJ mol^{-1}. Δ_o for $[Fe(H_2O)_6]^{2+}$ is about 125 kJ mol^{-1} and for $[Fe(CN)_6]^{4-}$ about 395 kJ mol^{-1}. Fe: Z = 26.

Answer Q55–58 on the basis of the above information.

Q55. $[Fe(H_2O)_6]^{2+}$ is: [1]

(A) low spin, $t_{2g}^6e_g^0$ **(B)** low spin, $t_{2g}^3e_g^3$
(C) high spin, $t_{2g}^4e_g^2$ **(D)** high spin, $t_{2g}^6e_g^0$

Answer: (C)
1. Δ_o < P ⇒ high spin d^6: $t_{2g}^4e_g^2$

Q56. The number of unpaired electrons in $[Fe(CN)_6]^{4-}$ is: [1]

(A) 2 **(B) 0** **(C)** 1 **(D)** 4

Answer: (B)
1. Δ_o > P ⇒ low spin t_{2g}^6 ⇒ 0

Q57. The magnetic moment of $[Fe(H_2O)_6]^{2+}$ is about: [1]

(A) 2.83 BM **(B)** 0 BM **(C)** 5.92 BM **(D) 4.90 BM**

Answer: (D)
1. n = 4 ⇒ √24 = 4.90 BM

Q58. The hybridisation of Fe in $[Fe(CN)_6]^{4-}$ (VBT) is: [1]

(A) sp^3d^2 **(B)** dsp^2 **(C) d^2sp^3** **(D)** sp^3

Answer: (C)
1. Inner d orbitals available after pairing ⇒ d^2sp^3

Section E

5-mark level (Long Answer type) · 2 × 5 = 10 marks · −1.25 per wrong

Q59. Draw the structures of the geometrical isomers of $[Co(NH_3)_4Cl_2]^+$ and $[Co(NH_3)_3Cl_3]$, and state which (if any) are optically active. The correct description is: Isomerism *CBSE PYQ* [5]

(A) $[Co(NH_3)_4Cl_2]^+$: cis and trans; $[Co(NH_3)_3Cl_3]$: only the mer form exists

(B) $[Co(NH_3)_4Cl_2]^+$: cis and trans; $[Co(NH_3)_3Cl_3]$: fac and mer; none is optically active

(C) $[Co(NH_3)_4Cl_2]^+$: cis and trans, cis is optically active; $[Co(NH_3)_3Cl_3]$: fac and mer

(D) $[Co(NH_3)_4Cl_2]^+$: only one form; $[Co(NH_3)_3Cl_3]$: fac and mer, both optically active

Answer: (B)

1. $[Co(NH_3)_4Cl_2]^+$ (MA_4B_2): cis (Cl–Co–Cl = 90°) and trans (180°)
2. $[Co(NH_3)_3Cl_3]$ (MA_3B_3): fac (three Cl on one face) and mer (three Cl in a meridian)
3. All four structures possess a plane of symmetry because only monodentate ligands are present
4. Hence none is optically active; optical activity in octahedral complexes needs a chiral arrangement such as cis-$[M(AA)_2B_2]$

Common trap: cis-$[Co(NH_3)_4Cl_2]^+$ is NOT chiral; only cis complexes with bidentate ligands like en are.

Q60. For the complex $[Fe(en)_2Cl_2]Cl$, identify (i) the oxidation number of iron, (ii) the hybrid orbitals and shape of the complex, (iii) its magnetic behaviour (assume en makes it low spin), (iv) the number of its geometrical isomers, (v) whether there is an optical isomer, and (vi) its IUPAC name. Bonding (VBT & CFT), Colour & Magnetism *CBSE PYQ (NCERT)* [5]

(A) (i) +3; (ii) d^2sp^3, octahedral; (iii) paramagnetic, 1 unpaired e^-; (iv) 2; (v) yes, the cis form; (vi) dichloridobis(ethane-1,2-diamine)iron(III) chloride

(B) (i) +3; (ii) dsp^2, square planar; (iii) paramagnetic, 1 unpaired e^-; (iv) 2; (v) yes, the cis form; (vi) dichloridobis(ethane-1,2-diamine)iron(III) chloride

(C) (i) +3; (ii) d^2sp^3, octahedral; (iii) diamagnetic; (iv) 3; (v) no; (vi) bis(ethane-1,2-diamine)dichloridoferrate(III) chloride

(D) (i) +2; (ii) sp^3d^2, octahedral; (iii) paramagnetic, 5 unpaired e^-; (iv) 2; (v) yes, the trans form; (vi) dichloridobis(ethane-1,2-diamine)iron(II) chloride

Answer: (A)

1. (i) $x + 2(0) + 2(-1) = +1 \Rightarrow x = +3$
2. (ii) Fe^{3+} $3d^5$; low spin ⇒ two 3d orbitals free ⇒ d^2sp^3, octahedral
3. (iii) Low spin d^5: t_{2g}^5 ⇒ 1 unpaired electron ⇒ paramagnetic ($\mu \approx 1.73$ BM)
4. (iv) Two geometrical isomers: cis and trans
5. (v) The cis isomer is optically active (exists as a d/l pair); trans is not
6. (vi) dichloridobis(ethane-1,2-diamine)iron(III) chloride (ligands alphabetical: chlorido before ethane-1,2-diamine)

◆ END OF ANSWER KEY ◆